Emergent Communication in Cooperative, Functionally Asymmetric, Multi-Agent Reinforcement Learning

Group 9
Open-Endedness and General Intelligence
University College London

Abstract

We study whether structured communication can emerge in cooperative multi-agent reinforcement learning under functional asymmetry. In our `TwoDoerBottleneckEnv`, a globally informed but non-acting *Seer* must guide two blind *Doers* through a corridor bottleneck and help them select private target objects. Communication is constrained through a discrete Finite Scalar Quantization (FSQ) bottleneck, making the learned protocol interpretable. Using MAPPO, we analyse the effects of communication bandwidth and curriculum learning on task success and Causal Influence of Communication (CIC). We find that training on the full task alone often leads to communication collapse, whereas a two-phase curriculum substantially improves convergence, final performance, and communication strength. Qualitative analysis further shows that the learned protocol is sparse and partially compositional. Overall, structured emergent communication arises from reinforcement learning alone, but depends strongly on training design and bottleneck size. Our code is publicly available at `https://github.com/BrianEzi/OpenEndedness`.

1 Introduction

1.1 Motivation

Cooperative multi-agent reinforcement learning (MARL) requires agents to coordinate behaviour, yet in many realistic settings agents have fundamentally different access to information. A robot with full sensor coverage cannot directly share its perception with a partner that operates blindly, and a planning agent that observes the full task cannot physically execute what it plans. Communication is the natural solution, but whether agents spontaneously develop structured, reusable communication protocols under such constraints remains an open research question (Foerster et al., 2016; Lazaridou and Baroni, 2020).

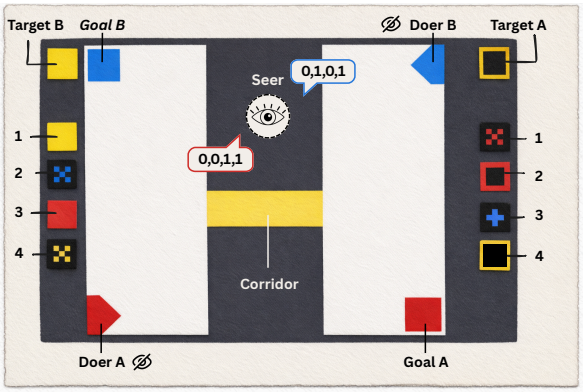

Figure 1: A Seer with global observability broadcasts discrete messages to two blind Doers (⊘), each of which must navigate through a three-cell corridor to reach its goal, then select its assigned target object from a shuffled 4-item menu (e.g., 1-4, drawn from a wider pool of 16 items).

A promising driver of structured communication is functional asymmetry, the deliberate separation of agents into roles defined by what they can perceive versus what they can do. When a globally-informed but action-restricted agent (*Seer*) must guide a perceptually-limited but action-capable partner (*Doer*), communication becomes the primary mechanism for task completion. This setup mirrors many real-world coordination systems, from air traffic controllers directing pilots to surgeons directing theatre teams. In each case, the informational hierarchy between roles creates pressure for efficient, consistent signalling conventions to emerge. We ask whether the same pressure, applied to artificial agents trained purely with reinforcement learning, is sufficient to induce language-like structure.

A second motivation concerns the nature of the communication channel itself. Continuous message vectors allow gradient flow, but they often lead to uninterpretable, partner-specific codes that do not generalise (Lewis et al., 2017). Discrete

communication channels are more linguistically meaningful, more efficient in bandwidth, and easier to interpret. A finite vocabulary of symbols can be logged, counted, and decoded. Discrete channels also connect to theories of human language origin, specifically the hypothesis that compositionality arose not from expressive need alone, but from the pressure to compress rich perceptual experience into learnable, transmissible signals (Eccles et al., 2019). We use Finite Scalar Quantization (FSQ) (Mentzer et al., 2023) as the communication bottleneck, forcing the Seer to compress its full global observation into a small vocabulary of reusable symbols.

Specifically, we study whether a stationary Seer agent with full global state access can develop a compositional discrete language that guides two perceptually limited, mobile Doer agents to complete tasks in a structured combinatorial space. The task requires navigating through a spatial bottleneck and selecting the correct item from a menu of 16 possible options (4 colours $\times$ 4 shapes), where each Doer holds a private target unknown to the other. We measure whether genuine causal communication emerges using the Causal Influence of Communication (CIC) metric (Eccles et al., 2019).

1.2 Research Hypothesis

We hypothesise that functional asymmetry in multi-agent tasks is sufficient to induce structured emergent communication via end-to-end reinforcement learning, without any explicit linguistic supervision. In particular, we predict that constraining a globally informed agent to communicate exclusively through a discrete bottleneck, while denying it the ability to act, creates sufficient pressure for that agent to develop a compressed, reusable symbol system that causally guides its action-capable but perceptually limited partners. Furthermore, we predict that this communication protocol can emerge through joint reinforcement learning alone, without requiring the informed agent to first master the task independently through pre-training.

1.3 Contributions

We make three main contributions: (i) we introduce a **functionally asymmetric environment** in which a Seer with global observability coordinates two blind Doers in a task that combines navigation through a bottleneck with combinatorial item selection; (ii) we demonstrate that, despite the complete absence of visual input for the Doers, a **discrete compositional communication protocol** reliably emerges that enables successful task completion; and (iii) we propose a **two-phase curriculum**, in which item selection precedes navigation, and show that this structured training regime substantially accelerates the emergence of effective communication.

2 Related Work

2.1 Emergent Communication in MARL

Emergent communication in multi-agent systems has been studied extensively since Foerster et al. (2016) showed that agents can learn to communicate via differentiable channels using deep RL. Their DIAL/RIAL framework established joint training of communicating agents as the standard approach, with both sender and receiver using similar recurrent architectures. Sukhbaatar et al. (2016) introduced CommNet, showing that agents sharing a homogeneous architecture and communicating via averaged message vectors could coordinate effectively in cooperative tasks. This homogeneous architecture design (i.e using the same backbone for all agents) has become standard practice (Kim et al., 2019), directly supporting our decision to use the same CNN+LSTM architecture for both Seer and Doer.

A key challenge in this literature is that successful coordination does not necessarily imply meaningful communication. Lewis et al. (2017) demonstrated that learned self-play communication protocols may be highly idiosyncratic, working only for specific agents trained together, and failing in cross-play scenarios. Mordatch and Abbeel (2018) addressed this through population-based training, encouraging more generalisable communication conventions.

2.2 Discrete Communication Channels

While continuous channels simplify optimisation, discrete communication offers superior interpretability and bandwidth efficiency (Müller et al., 2024; Chen et al., 2024). A standard framework for learning such representations is the Vector Quantised-Variational Autoencoder (VQ-VAE), which discretises latent spaces by mapping continuous vectors to their nearest neighbours in a learned codebook (van den Oord et al., 2017). However, VQ-VAE necessitates auxiliary training objectives, such as a commitment loss to stabilise the encoder and specific mechanisms for codebook updates,

which adds significant architectural complexity and hyperparameter sensitivity.

We instead utilise Finite Scalar Quantisation (FSQ), which discretises vectors by independently rounding each scalar dimension to the nearest integer within a fixed set of L levels (Mentzer et al., 2023). Unlike VQ-VAE, FSQ produces a discrete code from a fixed vocabulary of size L^d without requiring a learned codebook or commitment loss, as the rounding step is made differentiable via the straight-through estimator (Bengio et al., 2013). By forcing the Seer to communicate through this constrained discrete space, FSQ enhances protocol interpretability and ensures the experiments are relatable to a human linguistic context, which relies on discrete symbols rather than continuous signals.

2.3 Speaker-Listener Asymmetry

The most directly relevant work to our Seer-Doer setup is Karten et al. (2023), which introduces a "ghost agent" — a non-acting translator that mediates communication between trained agents and human partners. Their finding that the communicating agent should retain full access to task-relevant state directly supports our full-state Seer input design. Similarly, Lazaridou and Baroni (2020) emphasise that speakers in emergent communication settings are typically given the full meaning representation, while listeners must rely on the message.

This work motivates our use of functional asymmetry: the Seer can observe and communicate but not act, while the Doer can act but has restricted perception. Such asymmetry creates pragmatic pressure for communication to guide behaviour over time, rather than merely labelling observations. At the same time, Eccles et al. (2019) show that useful communication does not emerge automatically, and often depends on careful training design and inductive bias.

2.4 Evaluating Emergent Communication

Evaluating whether learned communication is meaningful remains a central challenge in the literature, as successful coordination does not inherently imply interpretable signalling. To monitor the functional utility of protocols, researchers often utilise the Causal Influence of Communication (CIC) score, which quantifies the degree to which messages causally impact a listener's action distribution. Post hoc analysis typically distinguishes between observational and interventional approaches: empirical rollout analysis provides a descriptive view of message usage within environmental contexts, while controlled codebook probes isolate the causal effect of specific messages by evaluating listener behaviour in fixed states. Furthermore, cross-play evaluation — pairing trained speakers with independently trained listeners — serves as a standard benchmark for determining whether learned protocols generalise into broader conventions rather than remaining idiosyncratic to a specific training pair (Lewis et al., 2017; Wolff et al., 2024). Beyond behavioural influence, structural analysis assesses the alignment between semantic and message spaces using measures such as topographical similarity (TopSim) (Lazaridou and Baroni, 2020). This evaluates whether the emergent protocol exhibits systematic grounding, where semantically similar states or targets are mapped to similar discrete signals, thereby reflecting a more human-like linguistic structure.

3 Method

3.1 Custom Two-Doer Bottleneck Environment

We implement a custom JAX gridworld, `TwoDoerBottleneckEnv`, designed to require coordination between two agents, as visualized in Figure 1. The environment is a 10×12 grid split into two rooms by a vertical wall, connected by a one-row, three-cell corridor that enforces coordinated passage. The global state is represented as a $(10, 12, 5)$ tensor encoding walls, agent positions, and goal locations, along with a symbolic state containing normalised positions, goal-reached flags, and step count.

At each episode start, Doer A and Doer B are placed randomly on opposite sides and must swap positions through the corridor. Upon reaching their goals, each agent receives a shuffled menu of four items drawn from a set of 16 (4 colours $\times$ 4 shapes) and must select its assigned target. The action space consists of nine discrete actions: movement (`stay`, `up`, `right`, `down`, `left`) and four pick actions, which are masked until the goal is reached. Episodes terminate upon correct selection by both agents, incorrect exhaustion of picks, or after 48 steps.

3.2 Architecture

We use MAPPO (Yu et al., 2022) with centralised training and decentralised execution: a shared critic observes the full state; each agent acts on local ob-

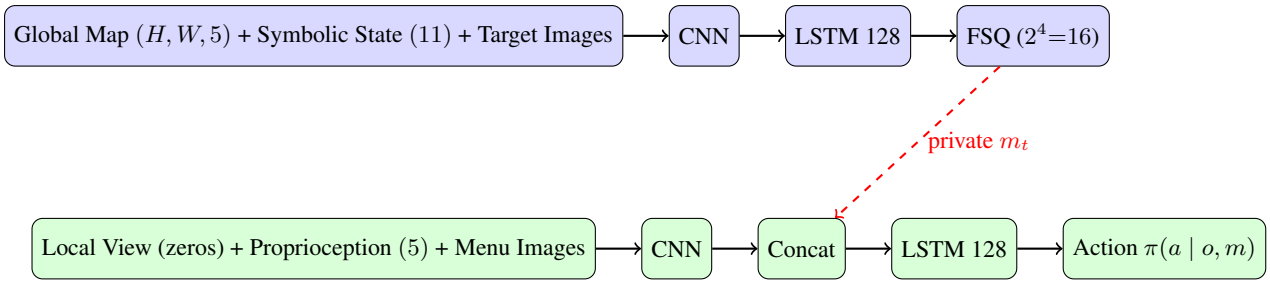

Figure 2: Seer-Doer architecture. The Seer encodes the full global state and sends a private discrete FSQ message to each Doer. The message is concatenated with the Doer's local features before the LSTM. LSTM 128 denotes an LSTM with hidden state dimension $h_t \in \mathbb{R}^{128}$.

servations. Communication is discretised via FSQ. We use d=4 dimensions at L=2 levels (equivalent to 4 bits), yielding a 16-symbol vocabulary. The full architecture is shown in Figure 2.

Seer. The Seer has no physical position in the grid. It receives the global map, an 11-dimensional symbolic state (normalised positions, goal-reached flags, step count), and both Doers' target item images (Table 1). These are encoded by separate CNN/MLP branches, concatenated, and passed through a 128-unit LSTM. The LSTM output is projected to $\mathbb{R}^d$ and quantised via FSQ to produce one private discrete message per Doer per timestep. An L2 penalty on the pre-quantisation vector prevents saturation and mode collapse.

Table 1: Seer inputs.

Input	Content	Shape
`global_map`	Walls, agent/goal positions	$(H, W, 5)$
`sym. state`	Positions, goals-reached, step	11
`target_img`	Target items per Doer	$(2, 5, 5, 3)$

Doers. Each Doer receives the inputs listed in Table 2: a zeroed-out 3×3 local view (ensuring dependence on the Seer's messages), a 5-dimensional proprioceptive vector (normalised position, one-hot agent identity, goal-reached flag), the private FSQ message, and menu images (zeroed until arrival at the goal). These are encoded by CNN/MLP/linear branches, concatenated, and processed by a 128-unit LSTM that outputs logits over 9 actions. Pick actions (5–8) are masked until the menu appears.

3.3 Training

All three agents are trained jointly. Separate Adam optimisers (lr $= 3 \times 10^{-4}$, $\varepsilon = 10^{-5}$, grad clip $= 0.5$) update the Seer, Doers, and critic. PPO uses $\varepsilon_{\text{clip}} = 0.2$.

Table 2: Doer inputs.

Input	Content	Shape
`local_view`	Grid view (zeros)	$(3, 3, 5)$
`propr.`	Position, identity, goal-reached	5
`message`	Private FSQ symbol	4
`menu_images`	4 options (at goal)	$(4, 5, 5, 3)$

Reward. The reward at timestep t is defined as

$$r_t = \begin{cases} +2.0 & \text{if both agents select correctly} \\ +0.5 & \text{if both agents reach their goals} \\ +0.5 & \text{per agent for correct selection} \\ -0.15 & \text{per agent for incorrect selection} \\ +0.1 \cdot \Delta d_t & \text{progress toward goal} \\ -0.03 & \text{per timestep} \\ -0.02 & \text{if hitting a wall} \\ -0.05 & \text{if collision occurs} \end{cases} \tag{1}$$

$$+ \alpha_{\text{cic}} \cdot \mathcal{L}_{\text{CIC}}, \tag{2}$$

where Δd_t is the reduction in Manhattan distance and $\alpha_{\text{cic}} = 0.01$. Here $\mathcal{L}_{\text{CIC}}$ is the empirical CIC score: the fraction of Doer actions that change when the Seer's messages are randomly shuffled across timesteps. This scalar is added as a positive reward bonus to incentivise informative communication.

Curriculum. The environment supports four perception levels: Level 0 provides full grid visibility; Level 1 local vision (3×3) and coordinates; Level 2 removes vision but retains coordinates; and Level 3 removes both. We train at Level 2 throughout: Doers know their position but cannot see their surroundings, forcing reliance on the Seer's messages. Levels 0 and 1 were excluded as even limited vision enables independent navigation, leading to commu-

nication collapse (Eccles et al., 2019). Level 3 was not used, as removing positional feedback slows convergence without adding meaningful pressure for communication. Full hyperparameters are listed in Table 7 in the Appendix.

4 Experiments

4.1 Experimental Setup

All experiments are conducted in the `TwoDoerBottleneckEnv`. To ensure that agents cannot solve the task through independent exploration and are forced to rely on the Seer's discrete signals, we utilise **Perception Level 2**. In this configuration, Doers' local 3×3 visual grids are zeroed out; each Doer's only inputs are its 5-dimensional proprioceptive vector (normalised position, one-hot agent identity, and goal-reached flag), the Seer's private message, and upon reaching the goal, its item menu. Training is performed using MAPPO across 16 parallel environments for up to 30 million timesteps. Each update collects 64 timesteps per environment (episodes terminate at 48 steps; unused buffer slots are masked), yielding 1,024 timesteps per update. Training Steps on all plots correspond to logging checkpoints every 10 updates; one Training Step thus equals 10,240 timesteps.

4.2 Experiment 1: Communication Bandwidth

We investigate the relationship between the available vocabulary size and the efficiency of the emergent protocol. Using the FSQ bottleneck, we vary the dimensionality of the discrete message space L^d:

- **3-bit ($2^3 = 8$ symbols):** A highly restrictive bottleneck testing the limits of signal compression.
- **4-bit ($2^4 = 16$ symbols):** The baseline capacity, corresponding to the levels $[2, 2, 2, 2]$.
- **5-bit ($2^5 = 32$ symbols):** A higher-capacity channel to determine if agents utilise increased expressive power or maintain a compressed protocol.

4.3 Experiment 2: The Role of Curriculum Learning

To test the hypothesis that communication emergence is sensitive to task complexity, we compare two training regimes:

- **Ablation (No Curriculum):** Agents are trained on the full task (Navigation and Item Selection) from the start with randomised initial positions.
- **Proposed (Two-Phase Curriculum):** Training begins with the `Pick_Object` phase, where Doers start at the goals to focus on decoding item-selection messages. Once first-try selection success exceeds 90% for three consecutive evaluations, the environment transitions to `Full_Training`, introducing the navigation bottleneck (see Table 7).

4.4 Evaluation Metrics

We assess performance using two primary metrics:

1. **Success Rate:** The percentage of episodes where both agents successfully navigate the bottleneck and select the correct items.
2. **Causal Influence of Communication (CIC):** Estimated as the mutual information $\mathcal{I}(A; M \mid S)$ under interventions on the message M, measuring how changes in the Seer's message affect the Doer's action distribution given the state S.

5 Results & Discussion

In the following, we analyse performance and communication via rollout success rate and CIC. Success requires both agents to reach their goals and correctly select their target items, while CIC measures the influence of messages on Doer actions. We then examine the learned protocol, focusing on message usage, semantics, and compositionality. A video visualisation of a successfully solved episode is available in our supplementary repository (`https://github.com/BrianEzi/OpenEndedness`).

5.1 Bandwidth Analysis

We analyse how communication bandwidth (i.e. how many bits the Seer can use to communicate) affects performance and the emergence of structured signalling. As shown in Figure 3, bandwidth induces a clear trade-off.

The 3-bit setting fails to converge, indicating insufficient capacity to encode both navigation and object identity. This is expected, since expressing 16 different colour-shape combinations combined with five navigation actions is very hard using only 3-bit. The 4-bit setting reaches stable mid-level

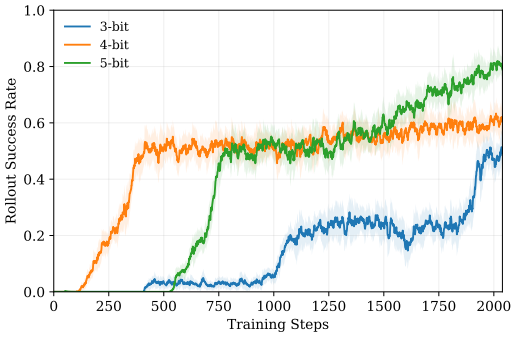

Figure 3: Rollout success rate across different Seer message bit-widths.

performance quickly, but does not manage to solve the task fully, converging around 60% success rate. In contrast, 5-bit Seer messages achieve up to 80% success rate but converge more slowly, due to a larger and harder-to-coordinate message space. Its stronger final performance is unsurprising because the 5-bit bandwidth is sufficient to learn a 1-to-1 mapping from message to all items and actions $(16+5)$, not forcing it to learn any compositionality in its language.

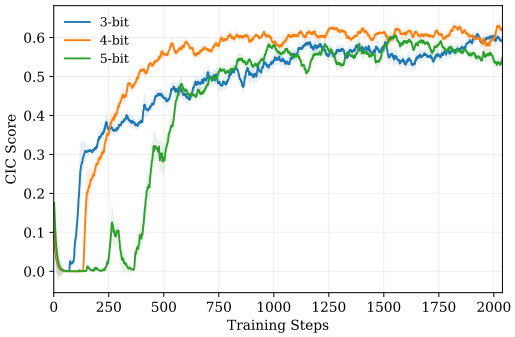

Figure 4: CIC Score across Seer message bit-widths.

Figure 4 shows that communication emerges in all settings, as the Seer's messages consistently influence Doer actions and thereby affect team performance. This is not a trivial task, because both Seer and Doers have not been pretrained before and therefore need to come up with their own communication code. However, different message bit-widths exhibit clear trade-offs: The 3-bit Seer message quickly reaches moderate CIC the quickest but plateaus, indicating limited expressivity. The 4-bit setting achieves the highest and most stable CIC, reflecting efficient and consistent communication. In contrast, 5-bit exhibits slower and noisier convergence, suggesting that larger bandwidth makes coordination harder.

5.2 Impact of Curriculum on Convergence

As described in Section 4.3, the curriculum first trains the agents on the `Pick_Object` phase before introducing full navigation. As shown in Figure 5, this leads to a characteristic two-stage learning dynamic. During the initial phase, performance improves rapidly as the agents learn to decode object-selection messages in isolation.

Upon transitioning to the full task, performance drops sharply, reflecting the sudden introduction of the navigation problem. However, the agents quickly recover and exhibit a steep improvement in success rate, surpassing all non-curriculum runs. This indicates that the previously learned communication protocol for object selection is retained and reused while learning navigation.

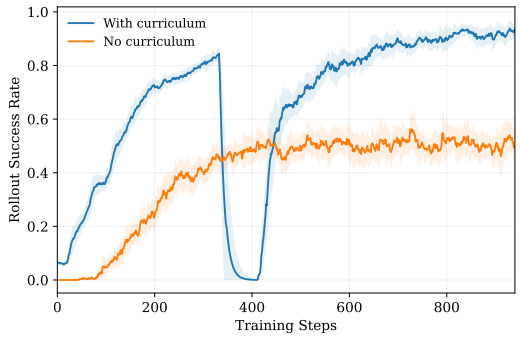

Figure 5: Rollout success rate in curriculum vs non-curriculum using 4-bit messages.

Notably, the agents do not relearn communication from scratch but extend the existing protocol to cover navigation. The same limited message set is reused across both phases, indicating polysemy, where messages take on context-dependent meanings despite the constrained 4-bit channel. Crucially, no catastrophic forgetting is observed, enabling rapid progression beyond the ∼50–60% performance ceiling seen without curriculum.

In contrast, the non-curriculum agent must learn navigation and selection jointly from sparse rewards, leading to slower progress and early plateauing without a consistent communication protocol.

This pattern is also reflected in the CIC scores in Figure 6. In the curriculum setting, CIC first increases during the object-selection phase, then drops sharply at the transition to the full task, and quickly recovers as the agents learn to incorporate navigation into the existing protocol. In contrast, without curriculum, CIC increases more gradually. While both settings eventually converge to similar CIC levels (∼0.6), the effectiveness of the resulting

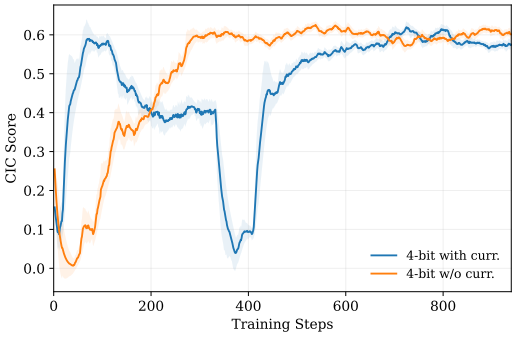

Figure 6: CIC scores indicating communication in curriculum vs. non-curriculum training.

protocols differs. The curriculum enables faster structuring and reuse of communication, whereas the non-curriculum setting requires slower joint optimization over a more complex space.

Protocol effectiveness is clearly reflected in the message usage distributions (Figure 7). The curriculum-trained model develops a specialised protocol anchored by Message ID 0, which functions as a dedicated "move forward" command. This high-frequency default allows the Seer to drive movement efficiently, diverging only for specific coordination or object selection. In contrast, the non-curriculum model exhibits a fragmented distribution lacking a dominant movement anchor, correlating with frequent navigation deadlocks. These results suggest that the curriculum enables a streamlined command structure by reserving specialised symbols for high-stakes coordination.

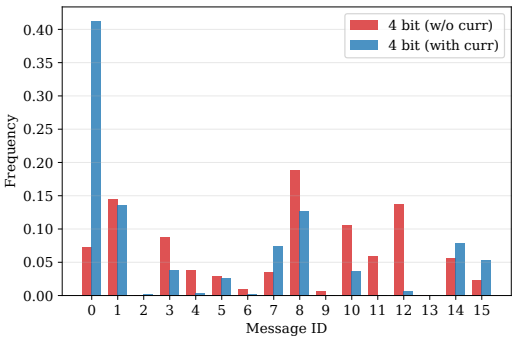

Figure 7: Message Frequency Distribution: A comparison of 4-bit models with and without curriculum.

5.3 Qualitative Analysis: Message Mapping

We analysed message–action correlations of our curriculum-trained policies from 1,000 greedy rollout episodes, recording the Seer's message at each Doer's first item selection as well as at every navigation step. Table 3 shows highly correlated message–action pairs for both phases. The full message-action breakdown can be found in appendix A.3.

Table 3: Message semantics for Doer A and Doer B across 1,000 episodes. Nav (A/B) is the dominant navigation action under each message (empirical purity $\geq$0.90 for Tier 1; †lower purity due to rare nav occurrence). N_{nav} is total navigation steps (A+B); N_{pick} is first-pick events. Item pur. is the fraction of picks matching the top item. Because Tier 1 messages have low item purity, the listed top item may vary across seeds; the key invariant is that navigation purity remains $\geq$0.90 and correct selection rate is 0.99.

Msg	Nav (A/B)	N_{nav}	N_{pick}	Top item	Item pur.	Col. pur.
Tier 1 — Navigation signals (low item purity, item resolved from LSTM context)						
`[1,0,0,0]`	down / down	2,653	707	yel. solid_sq.	0.24	0.49
`[0,0,0,1]`	up / up	2,178	639	red_frame	0.26	0.47
`[0,0,0,0]`	right / left	9,378	401	yel. plus	0.41	0.44
Tier 2 — Arrival–selection codes (high item purity, readable from message alone)						
`[1,1,1,1]`	stay / stay	438	272	blue_solid_sq.	0.56	1.00
`[0,1,0,1]`	up / up	162	253	blue_plus	0.68	0.68
`[1,1,1,0]`	down / stay†	1,788	185	blue_x	0.96	0.96
`[1,0,1,0]`	down / down	774	87	grn. solid_sq.	1.00	1.00
`[0,1,1,1]`	up / up	1,381	31	blue_frame	0.97	0.97

Sparse, two-tier vocabulary. Although a 16-symbol vocabulary is available, the Seer uses only 8–11 distinct messages per Doer, leaving the rest unused, indicating learned compression to the minimal set required for task success.

Messages split into two functional tiers, as listed in Table 3. *Tier 1* comprises three high-frequency messages ($\sim$68% of pick events) that induce consistent directional navigation: `[0,0,0,0]` drives right/left (toward the corridor), `[1,0,0,0]` drives down, and `[0,0,0,1]` drives up, each with empirical purity above 0.90. Because these signals are shared across targets, item-level purity is low (0.24-0.41), yet correct selection remains high (0.99), as target identity is recovered from accumulated LSTM context rather than the final message, hinting at compositionality of the emerged language code.

Tier 2 consists of low-frequency messages associated with `stay` or inward movement at the goal, signalling arrival or waiting for the other agent to cross the corridor. These messages are target-specific, yielding high item purity (0.68–1.00) and acting as direct item codes. For instance, `[1,0,1,0]` maps to `green_solid_square` (100% purity) and `[1,1,1,0]` to `blue_x` (96%), both with perfect selection accuracy.

Contextual polysemy. A striking property of the protocol is that every message serves double duty. The same symbol that guides navigation direction

during travel is reinterpreted as an item code at the goal. This is possible because pick actions are masked during navigation and movement is locked after arrival, so the Doer's behaviour is phase-gated by the environment. The Seer exploits this by using Tier 1 navigation messages to move the Doer efficiently and then switching to Tier 2 arrival–selection codes at the goal. The Doer's LSTM bridges the two phases, retaining target information encoded in early messages and activating the correct pick once the menu becomes visible.

Notably, for the same message, Doer A and Doer B consistently take mirrored or opposite navigation actions (`right` vs. `left` for `[0,0,0,0]`), reflecting their opposite starting positions. Item selections, however, are largely shared: `[1,0,1,0]` maps to `green_solid_square` for both Doers. The Seer therefore uses the private channel to personalise spatial guidance while broadcasting a common item vocabulary across both agents.

Partial colour compositionality. Bit-level analysis (Table 4) shows that bits 1 and 2 jointly and redundantly encode colour: when either is set, 83–87% of selections are blue items. Bit 3 provides a weak shape signal, loosely separating `x`-shaped from `frame`-shaped items (purity 0.48–0.52). No clean factorisation into independent colour and shape dimensions exists, but colour is genuinely distributed across bits — a hallmark of partial compositionality. Shape encoding remains weak, suggesting the 4-bit bottleneck exerts insufficient pressure to force a fully structured symbol system.

Table 4: Bit-level compositionality for Doer A. Purity is the fraction of picks matching the dominant colour or shape for each bit value. Bits 1 and 2 reliably encode blue vs. non-blue; bit 3 provides a weak shape cue; bit 0 contributes minimally.

Bit	Val.	Dom. col.	Col. pur.	Dom. shape	Shape pur.
0	0	red	0.36	plus	0.50
0	1	blue	0.36	solid_sq.	0.48
1	0	yellow	0.37	plus	0.27
1	1	blue	**0.87**	frame	0.32
2	0	yellow	0.33	plus	0.33
2	1	blue	**0.83**	solid_sq.	0.43
3	0	yellow	0.37	x	0.48
3	1	blue	0.40	frame	0.52

6 Conclusion

In this work, we investigated the emergence of communication protocols in a perception-constrained multi-agent environment using a Seer-Doer architecture. Our results demonstrate that the transition from independent navigation to coordinated action is highly sensitive to both task complexity and communication bandwidth.

We found that a two-phase curriculum is essential for avoiding communication collapse; by first grounding symbols in an object-selection task, the Seer and Doers can more effectively solve the subsequent navigation deadlock. Furthermore, our experiments with FSQ bottlenecks revealed that while a 4-bit channel provides an optimal balance for this task, extreme constraints (3-bit) force protocol compression, whereas higher bandwidth (5-bit) does not necessarily improve convergence speed. Ultimately, this study highlights the efficacy of treating communication as a learned information bottleneck that can be shaped through environmental pressure and structured training regimes.

Future Work

While our Seer-Doer architecture successfully solves the bottleneck task, several avenues for extension remain.

Generalisation to Dynamic Environments. Future work could evaluate the robustness of the learned communication protocol in procedurally generated maps or environments with moving obstacles, testing whether the vocabulary captures abstract structure or is overfitted to a fixed layout.

Hierarchical Communication. Introducing multi-level communication, where the Seer emits high-level intent signals and Doers exchange low-level coordination messages, could yield more structured protocols and better reflect hierarchical control in biological systems.

Zero-Shot Coordination. An important direction is to test whether Doers trained with a specific Seer can generalise to unseen Seers or even human operators using the same discrete vocabulary, providing a stronger measure of protocol generality.

Variable-Length Signalling. Moving beyond fixed-length FSQ codes to variable-length communication could allow agents to optimise bandwidth, using short messages for common actions and longer ones for rare or complex instructions.

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A Appendix

A.1 Architectural and Algorithmic Design Choices

Table 5: Key architectural and training decisions with supporting literature.

Decision	Choice	Literature Justification
Seer position	Global entity, no position in game	Karten et al. (2023)
Seer input	Full global map + symbolic state + target images	Lazaridou and Baroni (2020), Li and Zhang (2024)
Architecture	CNN + LSTM for both agents	Foerster et al. (2016), Sukhbaatar et al. (2016)
Message injection	After Doer CNN, before LSTM	Foerster et al. (2016), Sukhbaatar et al. (2016)
Training	Joint MAPPO from scratch	Yu et al. (2022), Foerster et al. (2016)
Vocabulary	$[2, 2, 2, 2]$ FSQ levels (vocab $= 16$)	Müller et al. (2024), Lazaridou and Baroni (2020)
Perception start	Level 2 (not 0/1) to prevent collapse	Eccles et al. (2019)
Evaluation	CIC + task success + message analysis	Eccles et al. (2019)

A.2 Environment Setup and Hyperparameters

Event	Reward
Both agents select correctly (task completion)	$+2.0$
Both agents arrive at goal (team, one-time)	$+0.5$
Individual correct selection	$+0.5$
Wrong selection	-0.15
Progress toward goal (per Doer, per step)	$\Delta d \times 0.1$
Step penalty	-0.03
Wall hit	-0.02
Agent collision	-0.05

Table 6: Reward structure of the `TwoDoerBottleneckEnv`.

Hyperparameter	Value
Total timesteps	30,000,000
Parallel environments	16
Steps per update	64
Episode max steps	48
LSTM hidden dim	128
Learning rate	3×10^{-4}
Adam ε	10^{-5}
Gradient clip norm	0.5
PPO clip ε	0.2
FSQ levels	$[2, 2, 2, 2]$
Vocabulary size	16
Seer entropy coef	0.05
CIC coef	0.01
Grid size	10×12
Corridor length	3
Selection advance threshold	90%
Curriculum success streak	3

Table 7: Hyperparameters used in the bottleneck experiments.

A.3 Extended Communication Analysis

The analysis of the communication protocols across different experimental conditions reveals how environmental constraints and training regimes shape emergent language. We evaluate this through vocabulary saturation and temporal stability.

Table 8: Full message lexicon across all 16 symbols. Nav action, purity, and N_{nav} are from empirical rollouts (1,000 episodes); indicates fewer than 5 steps observed. Top colour and shape are from empirical pick events; N_{pick} is first-pick count per Doer. Entries with $N_{\text{pick}} < 5$ marked * are unreliable. †Doer B picks `frame` instead of `solid_sq`.

	Doer A nav			Doer B nav						N_{pick}	
Msg	**Action**	**Pur.**	N_{nav}	**Action**	**Pur.**	N_{nav}	**Top colour**	**Top shape**	**Corr.**	**A**	**B**
`[0,0,0,0]`	right	0.98	5,053	left	1.00	4,325	red	plus	.99	81	118
`[0,0,0,1]`	up	0.90	1,067	up	0.74	1,111	red	frame	.98	124	86
`[0,0,1,0]`	—	—	0	—	—	0	—	—	—	0	0
`[0,0,1,1]`	up	1.00	133	up	0.96	870	—	—	—	0	0
`[0,1,0,0]`	—	—	0	stay	0.80	20	—	—	—	0	0
`[0,1,0,1]`	up	1.00	114	up	0.69	48	blue	plus	1.0	51	59
`[0,1,1,0]`	—	—	0	stay	0.94	54	—	—	—	0	0
`[0,1,1,1]`	up	1.00	548	up	0.89	833	blue	frame	1.0	10	3^*
`[1,0,0,0]`	down	0.99	1,189	down	0.90	1,464	yellow	x	.99	156	109
`[1,0,0,1]`	—	—	0	—	—	0	green	frame	1.0	2^*	0
`[1,0,1,0]`	down	1.00	165	down	1.00	609	green	solid_sq	1.0	19	0
`[1,0,1,1]`	—	—	0	—	—	0	—	—	—	0	0
`[1,1,0,0]`	down	1.00	4	stay	0.98	56	green	solid_sq	1.0	1^*	3^*
`[1,1,0,1]`	—	—	0	—	—	0	blue	frame	—	1^*	0
`[1,1,1,0]`	down	0.99	376	stay	0.58	1,412	blue	x	.99	42	56
`[1,1,1,1]`	stay	0.67	3	stay	1.00	435	blue	solid_sq†	1.0	50	78

A.3.1 Utilisation of Discrete Bandwidth

As shown in Figure 8, the Seer actively adapts its signalling strategy based on the available bit-depth. In the 3-bit condition, the vocabulary saturates almost immediately at its maximum of 8 symbols, suggesting that the task pressure exceeds the available expressive capacity. Conversely, the 5-bit model utilises a significantly larger vocabulary (approximately 25–27 unique symbols), indicating that the agents leverage the extra bandwidth to develop more nuanced instructions.

Interestingly, the 4-bit (with curriculum) model exhibits a distinct "discovery phase" between 100k and 400k timesteps, where the number of unique messages dips before stabilising at the maximum capacity of 16 symbols. This corresponds to the transition from the `Pick_Object` phase to `Full_Training`, where the agents must consolidate their object-naming vocabulary with new navigational requirements.

A.3.2 Temporal Evolution of the Protocol

The temporal heatmaps in Figure 9 provide a granular view of how the curriculum influences the internal structure of the emergent protocol.

- **Curriculum-Induced Specialisation:** In Figure 9a, a dramatic shift in message usage is visible at the 400k timestep mark. During the initial `Pick_Object` phase, the Seer utilises a specific set of signals (notably IDs 1, 14, and 15) to identify targets. Upon the introduction of the full task, Message ID 0 rapidly emerges as the primary signal. Qualitative analysis identifies ID 0 as a dedicated "move forward" command for both Doers. By establishing this high-frequency directional pulse, the Seer provides a consistent behavioural anchor that simplifies the navigation problem for the Doers.

- **Protocol Fragmentation:** In contrast, the non-curriculum heatmap (Figure 9b) reveals a lack of structural convergence. Without the initial grounding phase, the Seer never establishes a dominant movement anchor. Instead, the vocabulary is used in a fragmented manner, with multiple messages (IDs 3, 8, 11, and 12) competing for frequency without achieving functional stability. This fragmentation directly correlates with the lower success rates and frequent navigation deadlocks observed

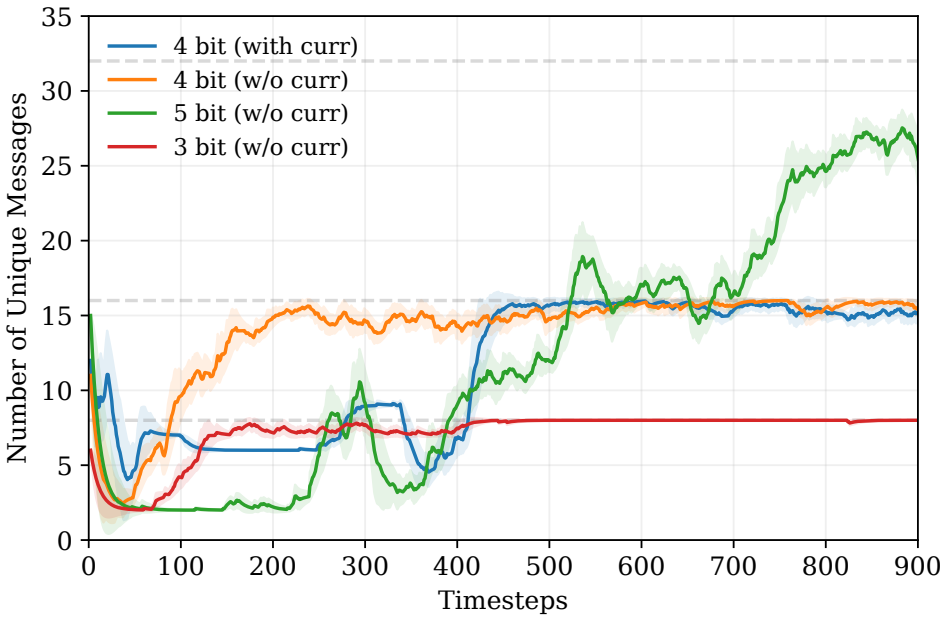
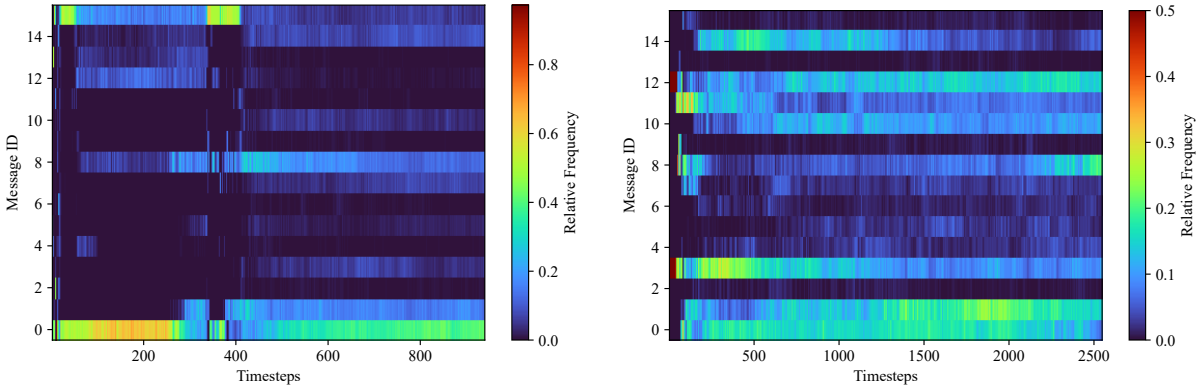


Figure 8: Vocabulary Saturation: Number of unique discrete messages utilised by the Seer over 900k timesteps. Models with higher bit-depth (5-bit) utilise nearly double the vocabulary of the 4-bit baseline, while the 3-bit model reaches capacity saturation early in training.

in non-curriculum runs; the absence of a clear, high-frequency command like the "move forward" signal in the curriculum model results in less efficient coordination.

This comparison suggests that the curriculum fundamentally alters the topology of the learned language. By first grounding symbols in a simpler selection task, the agents can effectively build an efficient command structure that uses a dominant signal for standard movement while reserving the remaining vocabulary for specific, high-stakes coordination and selection.



(a) With curriculum. A clear phase transition occurs at step ∼400k as the task shifts to full navigation. Message ID 0 stabilises as a dominant command, while other IDs remain specialised for selection.

(b) Without curriculum. The distribution remains fragmented throughout training. No single message emerges as a dominant anchor, leading to coordination instability.

Figure 9: Temporal heatmaps showing relative message frequency over training timesteps for 4-bit agents.